

ETF Universe Documentation

Generated: 2026-04-10

Total coverage: **4 markets, 309 ETFs across 14 thematic categories**

Overview

Market	Config File	Benchmark	ETFs	Themes	Data Dir	yfinance Suffix
Hong Kong	etf_config.json	HSI (Hang Seng Index)	74	10	data/	none (e.g. 02820.HK)
A-Share (CN)	a_share_etf_config.json	000300 (CSI 300)	136	13	data_ashare/	.SS / .SZ
Taiwan	tw_etf_config.json	0050 (TAIEX proxy)	27	7	data_tw/	.TW
South Korea	sk_etf_config.json	069500 (KOSPI 200)	72	10	data_sk/	.KS

Sources

Market	Primary Source	Notes
HK	yfinance (.HK suffix)	Good coverage; some dual-currency ETFs (HKD/USD/RMB)
CN	AkShare (fund_etf_hist_em)	yfinance works for most; AkShare needed for STAR Market codes (588xxx)
TW	yfinance (.TW suffix)	Sourced from investing.com/etfs/taiwan-etfs
SK	yfinance (.KS suffix)	Sourced from investing.com/etfs/south-korea-etfs

Theme Coverage by Market

Theme	HK	CN	TW	SK
semiconductor	5	12	5	7
technology_innovations	24	8	3	8
cloud_computing_robotics_ai	4	9	3	8
internet_communications	4	8	6	8
esg_clean_energy	17	11	5	8
biotech_healthcare	4	14	—	8
consumer_e_commerce	8	8	2	8
military_defense	—	4	—	8

Theme	HK	CN	TW	SK
real_estate	2	38	3	8
transportation	3	—	—	1
virtual_assets	3	—	—	—
broad_market	—	12	—	—
financials	—	5	—	—
gold_commodities	—	2	—	—
materials	—	5	—	—

Market Details

Hong Kong (HK)

Benchmark: Hang Seng Index (HSI)
Exchange: HKEX
Currencies covered: HKD, USD, RMB (dual/triple-listed ETFs included)
Sources: hkex.com.hk/

- Notes:**
- Many ETFs are dual-listed in HKD + USD (e.g. 02820 / 09820). Both are tracked; currency filter available in Industry Analysis page.
 - `virtual_assets` theme is unique to HK — includes Bitcoin and Ethereum spot ETFs approved in 2024 (03042, 03046, 03008).
 - `technology_innovations` is the largest theme (24 ETFs) due to broad HK tech ETF ecosystem.
 - Source: manually curated from HKEX ETF listings and Global X, ChinaAMC, CSOP issuer pages.

ETFs requiring attention:

- 03174 (CSOP China Healthcare Disruption) — underwent a stock split in 2025; split adjustment applied in `utils/config.py` via `STOCK_SPLITS`.

A-Share / China (CN)

Benchmark: CSI 300 (000300, tracked via AkShare)
Exchanges: SSE (Shanghai, 5xxxx/6xxxx) and SZSE (Shenzhen, 1xxxx/3xxxx)
Currency: CNY only
Sources: Excel sheet exported from etf.sse.com.cn

- Deduplication applied (2026-04-10):**
- Original config had 294 ETFs; reduced to **136** by:
- `broad_market`: 82 → 12 (kept 1 per major index: SSE50, CSI300, CSI500, CSI1000, A500, CSI Dividend, SSE180, CSI2000, HSI tracker, H-share tracker, S&P500; tracker)
 - `technology_innovations`: 34 → 8 (kept 1 per index: 科创50/100/200, 双创, HK Tech, CSI Tech, SOE Innovation)
 - `gold_commodities`: 14 → 2 (kept largest physical gold ETF 518880 + gold equity ETF 517400)
 - `financials`: 17 → 5 (kept most distinct: securities broker, bank, broad financial, fintech)
 - `materials`: 11 → 5 (kept 1 per sub-sector: rare earth, coal, non-ferrous, steel, petrochemicals)
 - Dropped `bonds_fixed_income` (15 ETFs) — out of scope
 - Dropped other (17 ETFs) — offshore index trackers (Nikkei, NASDAQ) not relevant to CN thematic analysis

Notes:

- `real_estate` (38 ETFs) are individual C-REITs — infrastructure, logistics, residential, commercial. These are distinct assets, not fund duplicates. All retained.
- STAR Market codes (588xxx) require AkShare for data collection — yfinance returns empty for these.
- Scraping logic: yfinance first → AkShare fallback; skip on rate limit.

ETFs requiring attention:

- All 588xxx codes (STAR Market) — yfinance cannot fetch these; AkShare `fund_etf_hist_em` required.
- 508xxx codes (C-REITs) — traded on SSE/SZSE but structured differently; verify data availability.

Taiwan (TW)

Benchmark: 0050.TW — Yuanta P-shares Taiwan Top 50 (largest TAIEX-tracking ETF, ~60% TSMC weight)

Exchange: TWSE / TPEX

Currency: TWD

Source: investing.com/etfs/taiwan-etfs

Exclusions applied:

- Symbols ending in L/R (leveraged/inverse)
- Symbols ending in U (commodity futures ETNs)
- Symbols ending in B (bond ETFs)
- Symbols ending in K (currency-hedged duplicates with negligible volume)

Notes:

- TW has a large dividend ETF ecosystem (00878, 00919, 0056) — excluded from thematic config as they are factor strategies, not sector/theme.
- Semiconductor theme is TSMC-heavy by nature of the Taiwanese market — all semiconductor ETFs have significant TSMC exposure.
- 00881 (Cathay Taiwan 5G Plus) is the most traded pure 5G/telecom ETF in TW.

ETFs requiring attention:

- 020000.TW (Fubon Big Apple Total Return Index) — tracks NYC-listed companies; unusual, verify yfinance coverage.
- 006201.TW (Yuanta GreTai 50) — TPEX-listed; confirm .TW suffix works in yfinance.

Major issuers: Yuanta (元大), Cathay (國泰), Fubon (富邦), CTBC (中信), Capital (群益), Fuh Hwa (富邦華一), SinoPac (永豐)

South Korea (SK)

Benchmark: 069500.KS — Samsung KODEX 200 (KOSPI 200, most liquid SK ETF)

Exchange: KRX

Currency: KRW

Source: investing.com/etfs/south-korea-etfs

Exclusions applied:

- Symbols starting with Q (ETNs, not ETFs)
- Leveraged/inverse products filtered by name keywords
- Bond/fixed-income products filtered by name keywords
- Capped at **8 ETFs per theme** (PDF order ≈ volume order, so top-8 are highest liquidity)

Notes:

- `esg_clean_energy` is dominated by secondary battery / EV ETFs — reflects Korea's role as a global battery manufacturer (Samsung SDI, LG Energy, SK Innovation).
- `military_defense` includes both K-Defense (국방) and Shipbuilding themes — Korea is a major defense exporter; these had significant inflows in 2025.
- `cloud_computing_robotics_ai` includes China Humanoid Robot ETFs (0048K0, 0053L0) — cross-market exposure.

- internet_communications is mostly NASDAQ 100 trackers — reflects Korean retail investor appetite for US tech.

ETFs requiring attention:

- Symbols with non-numeric characters (e.g. 0080G0, 0008T0, 0000Z0) — newer KRX codes; verify yfinance coverage with .KS suffix before scraping.
- 140710.KS (KODEX Transportation) — only 1 ETF in transportation theme; consider dropping theme or finding more.

Major issuers: Samsung (KODEX), Mirae Asset (TIGER), KB (KBSTAR/RISE), Shinhan (SOL), NH-Amundi (HANARO), Hanwha (ARIRANG), Timefolio, KIM (ACE)

Thematic Category Definitions

Theme Key	Description	Notes
semiconductor	Chip design, fabrication, EDA, equipment, materials	Includes upstream (equipment/materials) and downstream (fabless, IDM)
technology_innovations	Broad tech, innovation indexes, IT sector	Catch-all for tech ETFs not fitting narrower themes
cloud_computing_robotics_ai	AI, cloud, robotics, automation, humanoid	Fastest-growing theme across all markets in 2025
internet_communications	Internet platforms, 5G, telecom, NASDAQ	HK/CN focus on Chinese internet; TW/SK skew toward US NASDAQ
esg_clean_energy	Clean energy, EV, battery, ESG, carbon	SK dominated by battery ETFs; HK/CN by solar/wind
biotech_healthcare	Biotech, pharma, medical devices, healthcare	Includes cosmetics ETFs (SK: cosmetics classified here)
consumer_e_commerce	Consumer brands, retail, e-commerce, F&B;	HK/CN focus on China consumer; SK tracks global/US consumer
military_defense	Defense, aerospace, shipbuilding, UAV	CN: military SOEs; SK: K-Defense + shipbuilding
real_estate	REITs, property, infrastructure	CN: 38 individual C-REITs; HK/TW/SK: diversified REIT ETFs
transportation	Logistics, shipping, aviation	Limited coverage — HK has most (3), SK has 1
virtual_assets	Bitcoin/Ethereum spot ETFs	HK only — approved April 2024
broad_market	Major market indexes	CN only — SSE50, CSI300/500/1000, A500, etc.
financials	Banks, brokers, insurance	CN only
gold_commodities	Physical gold, gold mining, metals	CN only
materials	Rare earth, steel, coal, chemicals	CN only

Data Collection

```
# Activate environment
source venv/bin/activate

# Collect / update all market data
python data_collection.py # HK + CN (existing)
# TW and SK: to be added to data_collection.py
```

Scraping logic:

1. yfinance first (fast, no rate limit for HK/TW/SK)
2. AkShare fallback (CN STAR Market 588xxx codes only)
3. Skip gracefully on AkShare rate limit — do not block

Data directories:

Market	Directory	Format
HK	data/	data/{industry}/{code}.csv
CN	data_ashare/	data_ashare/{industry}/{code}.csv
TW	data_tw/	data_tw/{industry}/{code}.csv
SK	data_sk/	data_sk/{industry}/{code}.csv

Each CSV contains: Date, Open, High, Low, Close, Volume